

Saish Karole

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PROFESSIONAL SUMMARY

Electronics engineering undergraduate with hands-on experience in Computer Architecture, Deep Learning, and FPGA development. Research interests include Computer Architecture, AI-driven hardware design, memory systems, and learning-based methods for high-efficiency computing.

EDUCATION

Veermata Jijabai Technological Institute (VJTI)

Mumbai, India

B.Tech in Electronics Engineering | CGPA: 8.32 (Till Sem 6)

Nov. 2022 – Present

- Relevant Coursework: Digital Electronics, Microprocessors and Microcontrollers, Data Science, Deep Learning, Engineering Math.

SKILLS

Languages: C, C++, Embedded C, Python, Verilog HDL

HDL Tools: Yosys, Icarus Verilog, GTKWave, Quartus Prime, Vivado, Modelsim, Vitis IDE

Embedded Platforms: ESP-IDF, FreeRTOS, Raspberry Pi, Arduino, KV260

Developer Tools: Git/Github, Docker, CMake, Makefile, VS Code

Others: Linux, OpenCV, ScalaTest, Chisel HDL

EXPERIENCE

Aumnatic Systems Pvt Ltd

May 2025 – Jul 2025

Design Intern

Turbhe, Navi Mumbai

- Developed firmware for an ESP32-based prototype to test DC brushed motors under load.
- Implemented real-time voltage control, stall current monitoring, and data logging.
- Created a Python WebSocket client for remote control and real-time data visualization.

e-Yantra Robotics Competition (eYRC), IIT Bombay - Semi-finalists

Aug. 2023 – Mar. 2024

Team Member

Github

- Built a Pick and Place + Line Following robot simulating a space station task.
- Designed a single-cycle RISC-V RV32I core in Verilog to execute compiled C code.
- Ran Dijkstra's Algorithm on FPGA to compute shortest paths using a PID-controlled robot.
- Programmed on DE0-Nano (Cyclone IV FPGA) using Intel Quartus and Modelsim.

PROJECTS & RESEARCH WORK

Q-Cache

Oct. 2025 – Nov. 2025

Github Link

- Designed a Deep Reinforcement Learning-based cache replacement policy that learns how to evict cache blocks and compare its performance with traditional methods such as LRU, LFU, and Random
- Using a Deep Q-Network (DQN), the agent observes the cache state, predicts the long-term value of keeping each item, and chooses an eviction action that maximizes future hit rate.
- Evaluated Q-Cache across multiple synthetic access patterns such as Zipf, Loop, Bursty, Markov, Stride, and Mixed traces.

Research Implementation - CycleGANs

Aug. 2025 – Sep. 2025

Colab NB Link

- Re-implemented CycleGAN from the original paper (Zhu et al.) for unpaired image-to-image translation (Horses to Zebras and vice versa)
- Achieved realistic style transfer with preserved structure in training outputs after 45 epochs.

Matrix Multiplication on FPGA

Dec. 2024 – Feb. 2025

GitHub Link

- Implemented matrix multiplication in Verilog and in a C kernel for acceleration on KV260.
- Validated results via Vitis IDE and performance testing on Kria KV260 board.

Yosys Open-Source Contribution

Aug. 2024
GitHub PR #4546

- Merged a PR adding 23 new help messages for Verilog cells (e.g., \$shr, \$shl, \$add, \$mul, \$mux) in `techlibs/common/simlibs.v`
- Tested changes via Docker to improve usability and documentation for FPGA synthesis workflows

RISC-V RV32IM CPU on FPGA

Aug. 2023 – Oct. 2023
GitHub Link

- Designed and simulated a single-cycle RISC-V core in Verilog with testbenches.
- Implemented and verified instruction fetch, decode, and execute stages, ensuring correct arithmetic and branch operations.
- Deployed on UPduino 3.0 using Yosys + IceStorm; verified Fibonacci output on 7-segment display.

CERTIFICATIONS

Advanced Computer Architecture — NPTEL (IIT Delhi)

Elite + Silver — Top 5% of learners

2026
Credential

OTHER ACTIVITIES

uArch Workshop - ISCA'25, Tokyo

Attendee, selected with a full grant

June, 2025
Certificate

- Participated in sessions on cutting-edge research in Computer Architecture and engaged in professional networking with researchers and industry experts.

Society of Robotics and Automation (SRA), VJTI

Board Member

Aug. 2023 – Present
Website

- Organized robotics and embedded systems workshops for 220+ students.
- Conducted "Wall-E", "Pixels", and "MARIO" workshops on line following and self-balancing bots, computer vision, and micro-ROS.
- Mentored two FPGA-based project teams on Arty A7-35T under Eklavya Mentorship.